

0125700 | 宋友生.

1.

(a) $100/7 = 14$

$999/7 = 142 \Rightarrow 142 \cdot 7 = 994$

(b)

$1 \sim 100 : 50 \text{ odds.} \Rightarrow 450$

$1 \sim 999 : 500 \text{ odds}$

(c)

$111, 222, \dots, 999 \Rightarrow 9$

(d)

$1 \sim 100 : 25 \text{ divisible by } 4$

$1 \sim 999 : 249 \text{ divisible by } 4$

$\Rightarrow 100 \sim 999 : 224 \text{ divisible by } 4$

$\Rightarrow 900 - 224 \text{ not divisible by } 4.$

$\Rightarrow 676$

$$\begin{array}{r} 249 \\ 4 \overline{) 999} \\ \underline{199} \\ 199 \\ \underline{199} \\ 0 \end{array}$$

$$\begin{array}{r} 249 \\ 4 \overline{) 999} \\ \underline{199} \\ 199 \\ \underline{199} \\ 0 \end{array}$$

(e)

$$\begin{array}{r} 33 \\ 3 \overline{) 100} \\ \underline{99} \\ 1 \end{array}$$

$$\begin{array}{r} 8 \\ 12 \overline{) 96} \\ \underline{96} \\ 0 \end{array}$$

$$\begin{array}{r} 83 \\ 4 \overline{) 332} \\ \underline{332} \\ 0 \end{array}$$

$1 \sim 100 : 33 \text{ divisible by } 3, 24 \text{ divisible by } 4, 8 \text{ divisible by } 12$

$1 \sim 999 : 333 \text{ divisible by } 3, 249 \text{ divisible by } 4, 83 \text{ divisible by } 12$

$100 \sim 999 : 300 \text{ divisible by } 3, 225 \text{ divisible by } 4, 75 \text{ divisible by } 12.$

$\Rightarrow 300 + 225 - 75 = 450$

(f)

$900 - 450 = 450$

(g)

$300 - 75 = 225$

(h)

75

2.

鴿子: $\{1, 3, 5, 7, 9, 11, 13, 15\}$

鴿籠: $\{1, 15\}, \{3, 13\}, \{5, 11\}, \{7, 9\}$

鴿籠 (廣義): n 隻鴿子, m 個籠子,

必有至少一個籠子有 $\lceil \frac{n}{m} \rceil$ 隻

$$\Rightarrow \lceil \frac{x}{4} \rceil = 2 \Rightarrow x = 5$$

3. by 鴿籠:

(a) $\lceil \frac{x}{2} \rceil = 2, x = 3$

(b) 12 棕 2 黑, $x = 14$

4. (a)

$$\begin{aligned} & C_{10}^{10} + C_{3+5}^{10} + C_{5+7}^{10} + C_{7+9}^{10} + C_9^{10} \\ &= 10 + \frac{10 \times 9 \times 8}{3!} + \frac{10 \times 9 \times 8 \times 7 \times 6}{5!} + \frac{10 \times 9 \times 8}{3!} + 10 \\ &= 20 + 120 + 120 + 28 \times 9 \\ &= 240 + 20 + 252 = 512 \end{aligned}$$

(b)

$$\begin{aligned} & 2^{100} - C_0^{100} - C_1^{100} - C_2^{100} \\ &= 2^{100} - 1 - 100 - \frac{9900}{2} = 4950 \\ &= 2^{100} - 5051 \end{aligned}$$

$$5. (a) C_9^{19} \times 2^{10} \times (-x)^9$$

$$\Rightarrow \text{coefficient} : C_9^{19} \times 2^{10} \times -1$$

$$= -92378 \cdot 1024$$

$$= -94595072$$

(b)

$$C_9^{17} \times (3x)^8 \times (2y)^9$$

$$\Rightarrow \text{coefficient} : C_9^{17} \times 3^8 \times 2^9$$

$$= 24310 \cdot 3^8 \cdot 2^9$$

$$= 8166292920$$

6. Pascal's identity: $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

$$\binom{11}{1} = \binom{10}{0} + \binom{10}{1} = 1 + 10 = 11 \quad \binom{11}{0} = 1$$

$$\binom{11}{2} = \binom{10}{1} + \binom{10}{2} = 10 + 45 = 55$$

$$\binom{11}{3} = \binom{10}{2} + \binom{10}{3} = 45 + 120 = 165 \quad \binom{11}{11} = 1$$

$$\binom{11}{4} = \binom{10}{3} + \binom{10}{4} = 120 + 210 = 330$$

$$\binom{11}{5} = \binom{10}{4} + \binom{10}{5} = 210 + 252 = 462$$

$$\binom{11}{6} = \binom{10}{5} + \binom{10}{6} = 252 + 210 = 462$$

$$\binom{11}{7} = \binom{10}{6} + \binom{10}{7} = 210 + 120 = 330$$

$$\binom{11}{8} = \binom{10}{7} + \binom{10}{8} = 120 + 45 = 165$$

$$\binom{11}{9} = \binom{10}{8} + \binom{10}{9} = 45 + 10 = 55$$

$$\binom{11}{10} = \binom{10}{9} + \binom{10}{10} = 10 + 1 = 11$$

$$\Rightarrow 1 \quad 11 \quad 55 \quad 165 \quad 330 \quad 462 \quad 462 \quad 330 \quad 165 \quad 55 \quad 11 \quad 1$$

7.

$$(a) \quad X_1 \geq 1$$

$$\text{let } X_1' = X_1 - 1$$

$$\Rightarrow X_1' + X_2 + X_3 + X_4 + X_5 = X_1 + X_2 + X_3 + X_4 + X_5 - 1 = 20.$$



$$H_{20}^5 = C_{20}^{24} = \frac{24 \cdot 23 \cdot 22 \cdot 21}{4!} = 23 \cdot 22 \cdot 21 = 10626$$

$$\begin{array}{r} 23 \\ 22 \\ \hline 46 \\ 46 \\ \hline 506 \\ \times 21 \\ \hline 506 \\ 1012 \\ \hline 10626 \end{array}$$

$$(b) \quad X_i \geq 2 \text{ for } i = 1 \sim 5.$$

$$\text{let } X_i' = X_i - 2 \text{ for } i = 1 \sim 5$$

$$\Rightarrow X_1' + X_2' + X_3' + X_4' + X_5' = 21 - 2 \cdot 5 = 11.$$

$$H_{11}^5 = H_{11}^{15} = \frac{15 \cdot 14 \cdot 13 \cdot 12}{4!} = 7 \cdot 13 \cdot 15 = 1365$$

$$\begin{array}{r} 13 \\ \times 17 \\ \hline 91 \\ \times 15 \\ \hline 455 \\ 91 \\ \hline 1365 \end{array}$$

$$(c) \quad 0 \leq X_1 \leq 10 =$$

$$\Rightarrow \text{Total} - (X_1 \geq 11) \quad X_1' = X_1 - 11 \Rightarrow X_1' + X_2 + X_3 + X_4 + X_5 = 21 - 11 = 10.$$

$$H_{21}^5 - H_{10}^5 = C_{21}^{25} - C_{10}^{14} = \frac{25 \cdot 24 \cdot 23 \cdot 22}{4!} - \frac{14 \cdot 13 \cdot 12 \cdot 11}{4!}$$

$$= \frac{25 \cdot 24 \cdot 23 \cdot 22}{24} - \frac{14 \cdot 13 \cdot 12 \cdot 11}{24} = 25 \cdot 23 \cdot 22 - 7 \cdot 11 \cdot 13 = 12650 - 1001 = 11649$$

$$\begin{array}{r} 2 \\ 77 \\ \times 13 \\ \hline 231 \\ 77 \\ \hline 1001 \end{array}$$

7-

$$(d) \quad 0 \leq x_1 \leq 3 \\ 1 \leq x_2 < 4 \\ x_3 \geq 15.$$

$$\textcircled{1} \quad x_2 \geq 1 \text{ and } x_3 \geq 15$$

$$\textcircled{2} \quad x_1 \geq 4, x_2 \geq 1, x_3 \geq 15$$

$$\textcircled{3} \quad x_2 \geq 4, x_3 \geq 15$$

$$\textcircled{4} \quad x_1 \geq 4, x_2 \geq 5, x_3 \geq 15.$$

$$\textcircled{1} \quad x_2' = x_2 - 1 \\ x_3' = x_3 - 15.$$

$$x_1 + x_2' + x_3' + x_4 + x_5 = 21 - 16 = 5.$$

$$H_5^5 = C_5^9 = \frac{9 \times 8 \times 7 \times 6 \times 5}{4! \times 1} = 16 \times 7 = 112$$

$$\textcircled{2}$$

$$x_1' = x_1 - 4$$

$$x_2' = x_2 - 1$$

$$x_3' = x_3 - 15$$

$$x_1' + x_2' + x_3' + x_4 + x_5 = 21 - 20 = 1$$

$$H_1^5 = C_1^5 = 5$$

$$\textcircled{3}$$

$$x_2' = x_2 - 4$$

$$x_3' = x_3 - 15$$

$$x_1 + x_2' + x_3' + x_4 + x_5 = 21 - 19 = 2$$

$$H_2^5 = C_2^6 = \frac{30}{2} = 15$$

$$\textcircled{4}$$

$$4 + 5 + 15 > 21 \Rightarrow 0 \text{ solutions}$$

$$0 \leq x_1 \leq 3 \wedge 1 \leq x_2 \Rightarrow \textcircled{1} - \textcircled{2} = 112 - 5 = 107$$

$$0 \leq x_1 \leq 3 \wedge 1 \leq x_2 < 4 \Rightarrow \textcircled{1} - \textcircled{2} - \textcircled{3} = 112 - 15 = 97$$

8.

$$(a) a_0 = 1$$

$$a_1 = 1$$

$$a_2 = 2$$

$$a_3 = 3$$

$$a_4 = 5$$

$$a_5 = 8$$

$$\begin{array}{l} 1+1+1+1 \rightarrow 1 \\ 2+2 \rightarrow 1 \\ \textcircled{2+1+1} \rightarrow 3 \\ 1+2+1 \rightarrow 3 \end{array}$$

$$\begin{array}{l} 1+1+1+1+1 \rightarrow 1 \\ 2+2+1 \rightarrow 3 \\ 2+1+1+1 \rightarrow 4 \end{array}$$

$$a_n = a_{n-1} + a_{n-2}, \quad n \geq 2$$

b)

$$a_0 = 1$$

$$a_1 = 1$$

c)

$$a_6 = 13$$

$$a_7 = 21$$

$$a_8 = 34$$

9. (a) $a_n = 2a_{n-1}$ for $n \geq 1$, $a_0 = 3$.

Let $a_n = r^2$, $a_{n-1} = r$, $a_{n-2} = 1$

$$r^2 = 2r + 0$$

$$r^2 - 2r = 0$$

$$r(r-2) = 0$$

$$r = 0, 2$$

$$a_n = \alpha_1 \cdot 0^n + \alpha_2 \cdot 2^n$$

$$3 = a_0 = \alpha_2$$

$$\Rightarrow a_n = \alpha_2 \cdot 2^n = 3 \cdot 2^n$$

(b) $a_n = 5a_{n-1} - 6a_{n-2}$, $n \geq 2$

$$a_0 = 1$$

$$a_1 = 0$$

$$A: a_n = 3 \cdot 2^n - 2 \cdot 3^n$$

Let $a_n = r^2$, $a_{n-1} = r$, $a_{n-2} = 1$

$$r^2 = 5r - 6$$

$$r^2 - 5r + 6 = 0$$

$$r = 2$$

$$r = 3$$

$$r = 2 \text{ or } 3$$

$$a_n = \alpha_1 2^n + \alpha_2 3^n$$

$$2 = 2\alpha_1 + 2\alpha_2$$

$$1 = a_0 = \alpha_1 + \alpha_2$$

$$0 = a_1 = 2\alpha_1 + 3\alpha_2$$

$$\alpha_2 = -2$$

$$\alpha_1 = 3$$

9. (c)

$$a_n = \frac{1}{4} a_{n-2}, \quad n \geq 2. \quad a_0 = 1, a_1 = 0.$$

$$r^2 \cancel{r^n} = \frac{1}{4} \cancel{r^{n-2}}$$

$$\text{let } a_n = r^2, \quad a_{n-1} = r, \quad a_{n-2} = 1$$

$$r^2 = \frac{1}{4}$$

$$r = \frac{1}{2} \text{ or } -\frac{1}{2}$$

$$a_n = \alpha_1 \times \left(\frac{1}{2}\right)^n + \alpha_2 \times \left(-\frac{1}{2}\right)^n$$

$$1 = a_0 = \alpha_1 + \alpha_2$$

$$0 = a_1 = \frac{1}{2}\alpha_1 + \left(-\frac{1}{2}\alpha_2\right)$$

$$0 = \alpha_1 - \alpha_2,$$

$$2\alpha_1 = 1, \quad \alpha_1 = \frac{1}{2}, \quad \alpha_2 = -\frac{1}{2}$$

$$a_n = \frac{1}{2} \times \left(\frac{1}{2}\right)^n + \frac{1}{2} \left(-\frac{1}{2}\right)^n$$

$$= \left(\frac{1}{2}\right)^{n+1} - \left(-\frac{1}{2}\right)^{n+1}$$

9. (d)

$$a_n = 6a_{n-1} - 8a_{n-2} \text{ for } n \geq 2, a_0 = 4$$

$$a_1 = 10$$

$$\cancel{r^n} = 6 \cancel{r^{n-1}} - 8 \cancel{r^{n-2}}$$
$$r^2 = 6r - 8$$

$$\text{let } a_n = r^2, a_{n-1} = r, a_{n-2} = 1$$

$$r^2 = 6r - 8$$

$$r^2 - 6r + 8 = 0$$

$$\begin{matrix} r - 4 \\ r - 2 \end{matrix}$$

$$r = 4 \text{ or } 2$$

$$a_n = \alpha_1 \cdot 2^n + \alpha_2 \cdot 4^n$$

$$4 = a_0 = \alpha_1 + \alpha_2$$

$$10 = a_1 = 2\alpha_1 + 4\alpha_2$$

$$8 = 2\alpha_1 + 2\alpha_2$$

$$2\alpha_2 = 2, \alpha_2 = 1, \alpha_1 = 3$$

$$\Rightarrow a_n = 3 \cdot 2^n + 4^n$$

9. (e)

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}, \quad n \geq 3.$$

$$a_0 = 5, \quad a_1 = -9, \quad a_2 = 15.$$

$$\cancel{r^n} = -3\cancel{r^{n-1}} - 3\cancel{r^{n-2}} - \cancel{r^{n-3}}$$

$$r^3 = -3r^2 - 3r - 1$$

$$\text{let } a_n = r^3, \quad a_{n-1} = r^2, \quad a_{n-2} = r, \quad a_{n-3} = 1.$$

$$r^3 = -3r^2 - 3r - 1$$

$$r^3 + 3r^2 + 3r + 1 = 0.$$

$$(r+1)^3 = 0 \Rightarrow r = -1 \quad \text{3重根,}$$

$$a_n = \alpha_1 \cdot (-1)^n + \alpha_2 \cdot n(-1)^n + \alpha_3 \cdot n^2(-1)^n$$

$$5 = \alpha_1 \cdot 1$$

$$-9 = \alpha_1 \cdot (-1)^1 + \alpha_2 \cdot -1 + \alpha_3 \cdot -1^2$$

$$= -\alpha_1 - \alpha_2 - \alpha_3$$

$$15 = \alpha_1 + 2\alpha_2 + 4\alpha_3$$

$$\begin{array}{r} -8 \\ -4 \end{array} = \begin{array}{r} -\alpha_1 \\ -\alpha_2 - \alpha_3 \end{array}$$

$$10 = 2\alpha_2 + 4\alpha_3$$

$$2 = 2\alpha_3, \quad \alpha_3 = 1, \quad \alpha_2 = 3.$$

$$\Rightarrow a_n = 5(-1)^n + 3 \cdot n \cdot (-1)^n + n^2(-1)^n$$

$$10. (a) a_n = 2a_{n-1} + 2n^2$$

$$\text{let } a_n = r, a_{n-1} = 1$$

$$r = 2$$

$$a_n^{(h)} = \alpha \cdot 2^n$$

$$a_n^{(p)} = cn^2 + dn + e$$

$$cn^2 + dn + e = 2c(n-1)^2 + 2d(n-1) + 2e + 2n^2$$

$$= 2c(n^2 - 2n + 1) + 2dn - 2d + 2e + 2n^2$$

$$= (2c+2)n^2 + (-4c+2d)n + 2c-2d+2e$$

$$\Rightarrow \begin{matrix} (2c+2-C)n^2 + (-4c+2d-d)n + 2c-2d+e = 0 \\ C+2 \qquad \qquad \qquad d-4c \end{matrix}$$

$$c = -2, \quad d = 4c, \quad 2c - 2d + e = 0.$$

$$d = -8, \quad -4 + 16 + e = 0, \quad e = -12.$$

$$a_n^{(p)} = -2n^2 - 8n - 12$$

$$a_n = -2n^2 - 8n - 12 + \alpha(2)^n$$

(b)

$$a_1 = -2 - 8 - 12 + \alpha(2) = 4 \Rightarrow 2\alpha - 22 = 4$$

$$2\alpha = 26,$$

$$\Rightarrow a_n = -2n^2 - 8n - 12 + 13 \cdot 2^n$$

$$\alpha = 13$$

$$11. \quad a_n = 2a_{n-1} + 3^n.$$

$$(a) \quad \text{let } a_n = r, a_{n-1} = 1$$

$$r = 2.$$

$$\Rightarrow a_n^{(h)} = \alpha \cdot 2^n.$$

$$a_n^{(p)} = C \cdot 3^n$$

$$C \cdot 3^n = 2C \cdot 3^{n-1} + 3^n$$

$$C = 2C \cdot 3^{-1} + 1.$$

$$C = \frac{2}{3}C + 1$$

$$3C = 2C + 3.$$

$$C = 3 \Rightarrow a_n^{(p)} = 3 \cdot 3^n = 3^{n+1}$$

$$\Rightarrow a_n = \alpha \cdot 2^n + 3^{n+1}$$

$$(b) \quad a_1 = \alpha \cdot 2 + 3^2 = 5 \Rightarrow 9 + 2\alpha = 5 \Rightarrow \alpha = -2$$

$$\Rightarrow a_n = -2 \cdot 2^n + 3^{n+1}$$

$$= -2^{n+1} + 3^{n+1} \quad \#$$